Few-shot Adaptation Works with UnpredicTable Data

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Abstract

Prior work on language models (LMs) shows that training on a large number of diverse tasks improves few-shot learning (FSL) performance on new tasks. We take this to the extreme, automatically extracting 413,299 tasks from internet tables - orders of magnitude more than the next-largest public datasets. Finetuning on the resulting dataset leads to improved FSL performance on Natural Language Processing (NLP) tasks, but not proportionally to dataset scale. In fact, we find that narrow subsets of our dataset sometimes outperform more diverse datasets. For example, finetuning on software documentation from `support.google.com` raises FSL performance by a mean of +7.5% on 52 downstream tasks, which beats training on 40 human-curated NLP datasets (+6.7%). Finetuning on various narrow datasets leads to similar broad improvements across test tasks, suggesting that the gains are not from domain adaptation but adapting to FSL in general. We do not observe clear patterns between the datasets that lead to FSL gains, leaving open questions about why certain data helps with FSL.

1 Introduction

Brown et al. (2020) showed that language models (LMs) learn to perform new tasks from a few examples ("few-shot learning"; FSL). Explicitly training LMs for FSL further improves performance (Min et al., 2021; Chen et al., 2021b), and prior work has found that increasing the size and diversity of training tasks improves generalization to new tasks (Sanh et al., 2021; Aribandi et al., 2021; Aghajanyan et al., 2021a; Wang et al., 2022). We push size and diversity to the extreme by finetuning on a large dataset of automatically-curated FSL tasks, and surprisingly find that certain narrow datasets of tasks (e.g. software documentation) outperform much larger and more diverse datasets.

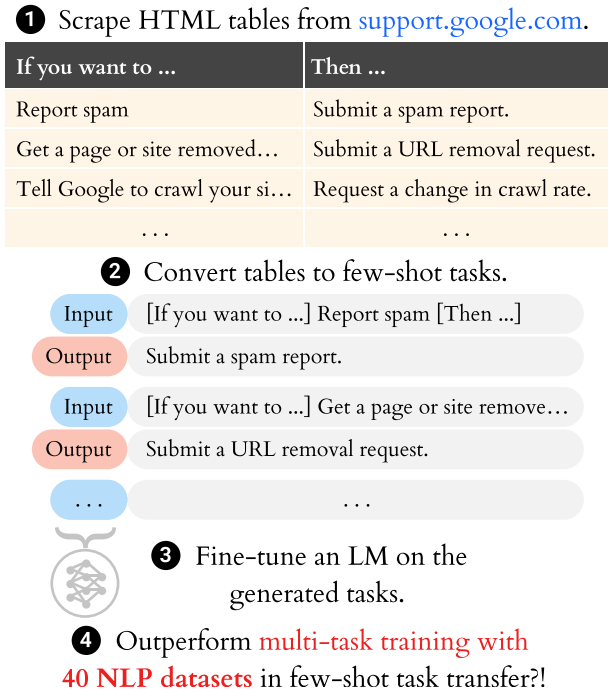


Figure 1: We convert web tables into FSL tasks, then use these tasks via finetuning to adapt language models for FSL. Unexpected tables lead to strong task transfer: finetuning GPT2 on software documentation from `support.google.com` outperforms finetuning on 40 curated NLP datasets on average across 52 test tasks, with strong improvements across diverse tasks including article classification (+47%), sentiment classification (+31%) and scientific question-answering (+23%).

Investigations into dataset size and diversity require a large dataset of FSL tasks. To this end, we explore tables as a naturally-occurring source of diverse FSL tasks. Given a table where each row is a list of fields, we hold out one row as the test example and treat all other rows as task training examples. We apply this idea to automatically convert internet tables into `UnpredicTable`[1], a dataset of 413,299 diverse few-shot tasks. We finetune GPT-2 to perform a new task given a few task examples in its context ("MetaICL"; Min et al.,

*Work done primarily at NYU and FAR.

[1]github.com/AnonCodeShare/few-shot-adaptation

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2021). Finetuning on UnpredicTable leads to strong FSL performance on average over 52 NLP test tasks. However, the observed gains fall short of expectations for such a large dataset.

To understand why our gains were limited, we perform ablations on dataset size, diversity, and content. We find that finetuning on narrow subsets of UnpredicTable outperforms finetuning on our diverse dataset and on curated NLP data. Surprisingly, datasets that we handpick according to what we expect to be helpful are not strongly correlated with performance. In fact, the training datasets that lead to strong improvements are often counterintuitive, covering trivia content (e.g. video games and software documentation; see Fig. 1) that are unrelated to test tasks. Finetuning on these narrow datasets cause broad improvements similar to finetuning on curated NLP datasets when compared on the same test tasks. This suggests that these aren't domain- or task-specific improvements, but improvements in general few-shot ability ("few-shot adaptation"). Our work calls into question common wisdom that adapting LMs to FSL requires diverse, high-quality training data.

2 Web Tables Are Few-Shot Tasks

We begin by describing FSL, which is the problem of learning from a small number of training examples. We make the case that web tables can be used as a diverse source of few-shot tasks. Then, we introduce our algorithm for converting tables into tasks and apply this to produce UnpredicTable, a dataset of 413,299 few-shot tasks.

2.1 Few-Shot Learning Tasks

We define a task T as a set of input-output pairs $T = \{(x_i, y_i)\}_{i=1}^{k}$ where inputs x_i map to outputs y_i. Tasks can be very diverse, from question-answering (Questions $\rightarrow$ Answers), to summarization (Books $\rightarrow$ Summaries), to translation (French $\rightarrow$ English). In FSL, k is small. LMs can be used to perform FSL by providing k training pairs $\{(x_i, y_i) : i = 1, \ldots, k\}$ in the LM context. Then, given a new example x_{target} for which y_{target} is unknown, we use the model to predict y_{target}.

2.2 Tables Dataset

Motivated by prior work on FSL adaptation (Min et al., 2021; Chen et al., 2021b) and multi-task learning (Sanh et al., 2021; Aribandi et al., 2021; Aghajanyan et al., 2021a), we hypothesize that we can extend the results of multi-task FSL finetuning with an even larger set of few-shot tasks. We make the case that web tables are a large and diverse source of few-shot tasks. Consider a table where each row is an instance of a similar class and columns describe the attributes of an instance. We use each row as an example of a task, where the task is filling in missing attributes in a row. For a table with k rows, each table becomes a k-shot dataset for a particular task.

As a source of table data, we use tables from the English-language Relational Subset of the WDC Web Table Corpus 2015 (WTC)[2] (Lehmberg et al., 2016). The WTC dataset was extracted from the July 2015 Common Crawl web corpus, and contains 50M tables from 323K web domains. We focus on relational tables, which describe a set of similar items along with their attributes. For example, a table listing national dishes by country is a relational table, while a table where each row describes a different attribute of a single item is not. WTC also provides helpful metadata including the source URL, title, and header rows.

2.3 Turning Tables Into Tasks

In practice, there are important design choices for converting a table into a task of input-output pairs. Here, we describe our chosen procedure. We start with the assumption that items in the relational table are listed row-wise (as in Fig. 2) instead of column-wise. Where necessary, we transpose the tables to suit our requirement. To convert a row into an input-output task pair, we consider a single column as a potential output target y_i and concatenate the remaining columns to form the input x_i. For additional context, we prefix each value with its column header (see Fig. 2). Since any column is a potential output target, we create multiple tasks per table. For example, a table with 3 columns A, B, and C may be cast as three different tasks: $P(A|B, C)$, $P(B|A, C)$ and $P(C|A, B)$. Exhaustively converting every column from every table into a new task leads to a large number of junk tasks, so we filter out tasks that do not meet basic criteria of task coherence (see Appendix A).

We apply our tables-to-tasks procedure to produce UnpredicTable, a dataset with 413,299 tasks from 23,744 websites. The shape of our dataset is different from most NLP datasets: NLP datasets typically contain a handful of

[2] webdatacommons.org/webtables/2015/EnglishStatistics

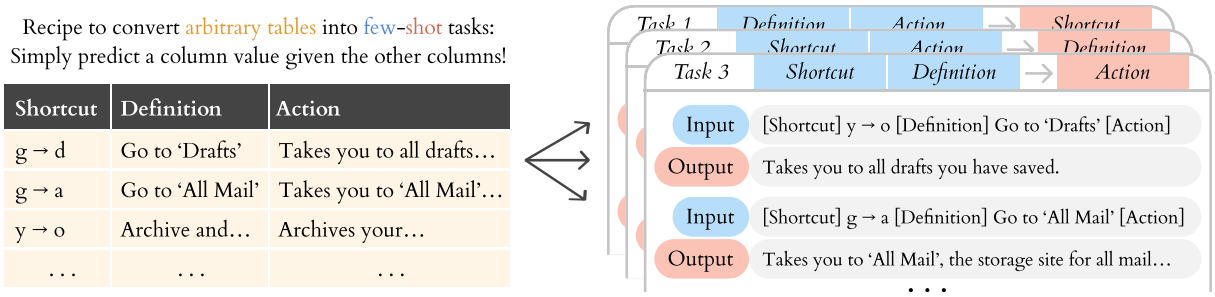


Figure 2: An algorithm to convert tables into tasks for FSL: Given the task of "Predict this column value given the other column values as input," each row in the table can be used as an example for that task.

tasks, with thousands of examples per task. `UnpredicTable` contains 400K tasks but most tasks have fewer than 50 examples. Thus, our dataset has a large variety of tasks but each task has limited training examples, true to the small-k FSL setting. Our code and dataset are open-source.[3]

3 Multitask Training with Few-shot Tasks for Few-shot Adaptation

The shape of our dataset makes it suitable for multitask learning algorithms. In multitask learning, we have a training dataset $\mathcal{D}_{\text{train}} = \{T_i\}_{i=1}^{M_{\text{train}}}$ containing M_{train} training tasks T, and a test dataset $\mathcal{D}_{\text{test}}$ with M_{test} tasks which are disjoint to $\mathcal{D}_{\text{train}}$. The key idea is to use $\mathcal{D}_{\text{train}}$ to train a model to be generalizable to new tasks in $\mathcal{D}_{\text{test}}$.

Here, we focus on the MetaICL algorithm (Min et al., 2021) for few-shot adaptation, which has shown strong FSL results across a variety of downstream tasks. To study the generalization of our results across different training algorithms, models and test tasks, we include additional experiments in Appendix D including zero-shot results and evaluation on the CrossFit (Ye et al., 2021) and FLEX (Bragg et al., 2021) benchmarks.

3.1 MetaICL

MetaICL (Min et al., 2021) trains LMs to predict the output for a target input, given a few input-output pairs provided in the LM context. On each training iteration, one task T_i is sampled from $\mathcal{D}_{\text{train}}$ and $k+1$ training examples $\{(x_1, y_1), \ldots, (x_{k+1}, y_{k+1})\}$ are sampled from T_i. MetaICL trains an LM with parameters θ to maximize $\log P(y_{k+1}|x_1, y_1, \ldots, x_k, y_k, x_{k+1})$. At test time, for a new task in $\mathcal{D}_{\text{test}}$ we draw a set of examples $\{x_1, y_1, \ldots, x_k, y_k\}$ and a query x_{k+1}. Given this context, the LM uses θ to select the most likely y_{k+1} from a discrete set of possible labels.

3.2 Experiments

Here, we investigate how finetuning on `UnpredicTable` compares to finetuning on human-curated NLP datasets. We finetune the 774M parameter pretrained GPT2-large LM (Radford et al., 2019), following Min et al. (2021). See Appendix C for details on our hyperparameter and finetuning setup.

NLP datasets and evaluation settings Min et al. (2021) use 142 unique NLP tasks from Ye et al. (2021) and Khashabi et al. (2020) to form $\mathcal{D}_{\text{train}}$ and $\mathcal{D}_{\text{test}}$ for 5 different NLP task categories: 26 *Low Resource* (LR) tasks with <1000 examples per task, 8 *Natural Language Inference* (NLI) tasks to test entailment between a premise and hypothesis clause, 4 *Paraphrase* (Para) tasks that test the equivalence of two differently-worded phrases, 20 *Classification* (Class) tasks, and 22 *Question-Answering* (QA) tasks. We show results on each category. See Appendix C for a full list of tasks.

MetaICL methods MetaICL evaluates performance on each task category in two ways. First, they consider an out of distribution ("OOD") setting, where they finetune a model on a dataset $\mathcal{D}_{\text{train}}$ consisting of tasks from all other categories excluding the target task category. Second, for *Class* and *QA* categories, they consider an in-domain ("IID") setting, where they finetune a model on a dataset $\mathcal{D}_{\text{train}}$ consisting of only tasks from the same category as the target task category.

Our dataset We sample $M = 5000$ tasks from `UnpredicTable`, choosing M based on results on a development set of tasks (Appendix C). We refer to this dataset as `UnpredicTable-5k`. Min et al. (2021) train one model per task category,

[3] github.com/AnonCodeShare/few-shot-adaptation

while we fine-tune a single GPT2-large model on `UnpredicTable-5k` and test the resulting model on all task categories.

3.3 Results

	Task category [# test tasks]				
Method	**LR**	**Class**	**QA**	**NLI**	**Para**
GPT2 0-shot	34.9	34.2	40.4	25.5	34.2
GPT2 k-shot	38.2	37.4	40.2	34	33.7
MetaICL k-shot trained with					
NLP (OOD)	43.2	38.2	38.7	**49**	33.1
NLP (IID)	-	43.4	**45.9**	-	-
UnpredicTable-5k (our dataset)	**43.7**	**46.1**	42.3	36.3	**45.7**

Table 1: Columns represent different test settings; rows represent different methods. *MetaICL k-shot* with finetuning on our dataset improves pretrained model performance (*GPT2 k-shot*) on all test categories. Furthermore, finetuning on our tasks beats finetuning on out-category NLP datasets (*OOD*) on 4/5 settings, and in-category NLP datasets (*IID*) on 1/2 settings.

For each category, we report the mean task accuracy for all tasks in the category. Tab. 1 shows the results. MetaICL finetuning on our table tasks improves FSL performance on all test settings. Furthermore, finetuning on our dataset outperforms finetuning on OOD NLP tasks on 4/5 settings, and IID NLP tasks on 1/2 settings. Overall, finetuning on our data results in comparable performance to finetuning on curated NLP tasks.

4 Why Is UnpredicTable Helpful?

To understand why `UnpredicTable` is helpful training data, we construct subsets of the dataset varying features we wish to study. For each sub-dataset, we finetune on that dataset individually following the setup as before (Appendix C) and measure FSL performance on MetaICL test tasks from all categories (52 total). All experiments are repeated for 3 random seeds to minimize the effects of random task sampling in each dataset. We report the mean accuracy from each experiment in Fig. 3.

4.1 Does increasing dataset size improve finetuning performance?

Fig. 3a shows FSL performance for differently-sized datasets randomly sampled from `UnpredicTable`. Each dataset has a maximum number of examples per task $N = 10$ and varies the number of tasks T. Increasing the number of tasks from $T = 40$ does not help and performance deteriorates beyond $T = 5000$, contrary to results in Wang et al. (2022).[4] Overall, the number of tasks does not seem to be the key factor for our finetuning transfer success.

4.2 Does diversity improve performance?

Next, we study the effect of task diversity on FSL performance. Tasks from the same website tend to be similar in content, so we construct more diverse datasets by sampling tasks from `UnpredicTable-unique`, a version of `UnpredicTable` filtered to have a maximum of one task per website (vs. up to 2500 in `UnpredicTable`). Fig. 3a shows that the difference between `UnpredicTable-unique` and `UnpredicTable` at matching sizes is small, suggesting that dataset diversity is not an important factor for our finetuning transfer success.

To examine narrow datasets in contrast to the uniformly-sampled ones, we consider 3 types of datasets grouped by content. We sample tasks from 20 websites of different genres, forming a dataset from each website (Fig. 3d). Secondly, we also form datasets of semantically similar tasks by clustering `UnpredicTable-unique` tasks into 30 clusters using HDBSCAN[5] (McInnes et al., 2017) (Fig. 3c). Finally, we also sample 20 NLP tasks from the 90 MetaICL training tasks and use each task as a separate training dataset (Fig. 3e). Single-website and single-NLP datasets have $T \times N = 10000$ total examples, and cluster datasets have different T due to the clustering algorithm.

We find significant variance among the narrow datasets. Some single-website or cluster datasets are better than diverse datasets, such as `support.google.com` which is our best dataset overall (even outperforming diverse NLP datasets). This suggests that diverse task datasets can be replaced with careful selection of a narrow training dataset for FSL improvement.

4.3 Can we select good tasks by hand?

Padmakumar et al. (2022) found that some training tasks can negatively impact downstream perfor-

[4] For additional dataset scaling results, we randomly sample human-curated NLP tasks from the MetaICL training set (Fig. 3b). Since there are only 90 NLP training tasks, we use $T = 40$ tasks and vary N to match the total number of examples in Fig. 3a. At an equal number of tasks and examples per task ($T = 40$, $N = 10$), NLP datasets outperform our dataset by $\sim 1\%$. (The results in Tab. 1 differ due to the choices of train and test tasks in different task categories.)

[5] See Appendix E for details of our clustering setup.

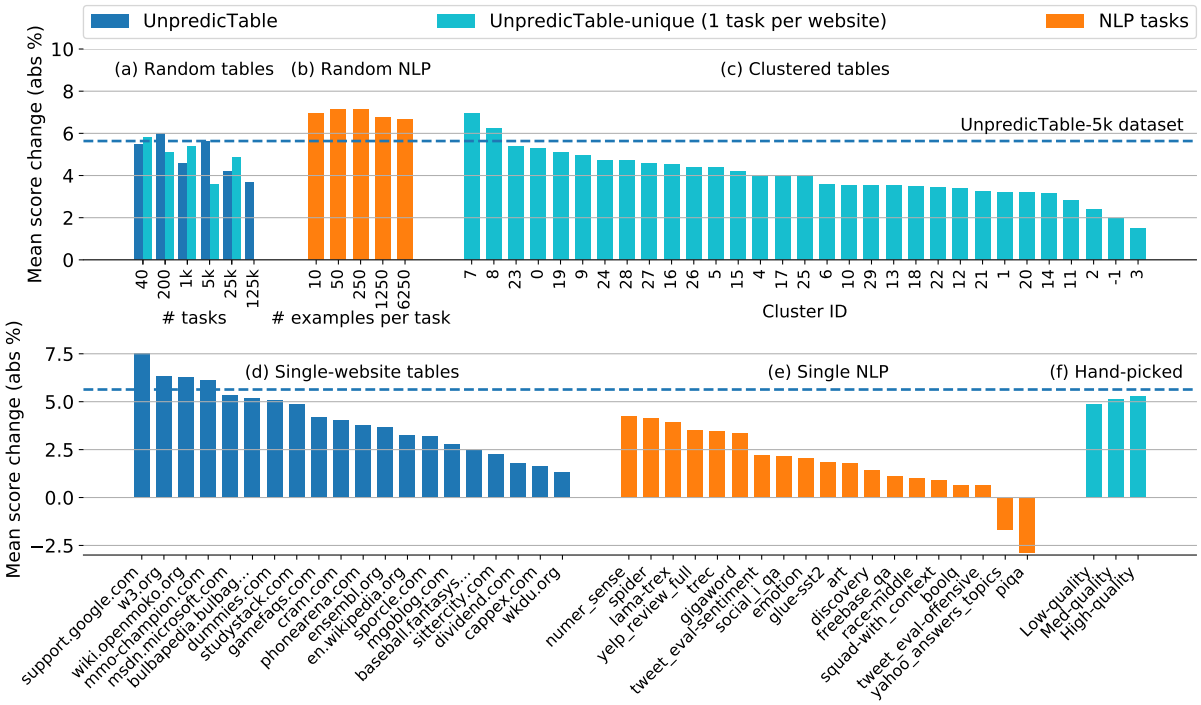


Figure 3: Each bar represents a GPT2 model finetuned on a different dataset. The y-axis shows mean improvement of a finetuned LM over the pretrained LM. **Comparing dataset helpfulness:** Datasets made of diverse tasks from UnpredicTable (a) and NLP datasets (b) lead to +5–7% improvement. Narrow clusters (c) and websites (d) within UnpredicTable vary significantly, with the best narrow datasets matching the best multi-task NLP datasets (b).

mance, which could explain why aggregating many random tasks may be less successful than individual tasks. We manually categorize 2,000 tasks from `UnpredicTable-unique` into High, Mid, and Low-quality.[6] We define low-quality tasks as tasks where the content is junk or relies on missing context. High-quality tasks are ones where an annotator could pick the correct answer from a list of options, and tests useful abilities (logic, general knowledge, comprehension, etc.). Mid-quality tasks are the remaining tasks. For each class, we randomly sample $T = 200$ tasks to form its own dataset.

Surprisingly, our manual annotations of quality are not strongly correlated with downstream task performance (Fig. 3f). Our handpicked dataset of high-quality tasks does not even surpass the scores of randomly-sampled tasks, and the difference in performance between our low and high-quality datasets are <1%. These results suggest that tasks that look helpful are not necessarily helpful.

4.4 How do helpful and unhelpful tasks look?

We look for features of helpful and unhelpful datasets with examples from cluster, single-website and single-NLP datasets. 4/5 of the most helpful datasets are software-related. `support.google.com`, `w3.org` and `wiki.openmoko.org` contain software documentation; `cluster 7` describes information related to internet cookies. Unhelpful datasets are more varied. The two least-helpful datasets are NLP datasets: `piqa` (question-answering task for physical knowledge) and `yahoo_answers_topics` (topic-classification task) both yield negative transfer results. The least helpful table datasets include highly-repetitive software tables (`cluster 2 & 3`), tasks classified as noise by the clustering algorithm (`cluster -1`), college review posts (`cappex.com`), and music database entries (`wkdu.org`).

The top datasets appear unrelated to our test tasks (e.g. there are no software-related test tasks). Additional examples highlight this: `mmo-champion.com` and `bulbapedia.bulbagarden.net` are video game trivia sites that do not seem useful for other tasks, yet these datasets are on par with `UnpredicTable-5k`. Conversely, websites containing high-quality question-answer pairs such as `cram.com` and `studystack.com`, as well as `en.wikipedia.org` which contains many

[6] See Appendix F for details of our annotation setup.

Examples of Helpful Tasks	
`w3.org`	
input	[Keyword] password [Data type] Text with no line breaks (sensitive information) [State]
output	Password
`bulbapedia.bulbagarden.net`	
input	[Move] Odor Sleuth [Effect]
output	Never ends, screen freezes with the words "Wild/Foe (Pokémon) used Odor Sleuth!"
`cluster 7`	
input	[Cookie] guest_id, ki [Information]
output	These cookies allow you to access the Twitter feed on the homepage.
Examples of Unhelpful Tasks	
`wkdu.org`	
input	[Artist] Noah and the Whale [Title]
output	5 Years Time
`cappex.com`	
input	[Comments] ... anything you would want to do is just an easy ten minute drive away. [Categories]
output	What to do for fun
`yahoo_answers_topics`	
input	bungee jumping site in victoria??? i am trying to find a site for bungee jumping ... *(Truncated)*
output	Sports

Table 2: Helpful and unhelpful datasets are highly varied and do not always match intuitions on task quality.

real-world facts, yield subpar improvements. We include examples of helpful and unhelpful tasks in Tab. 2, and more examples in Appendix G.

4.5 Which tasks are our datasets helpful for?

Here, we investigate which test tasks benefit from our finetuning. Fig 4 shows score improvements on all 52 test tasks relative to the pretrained model after finetuning on `UnpredicTable-5k`, `NLP-1250`[7], and `support.google.com`. Summary statistics are shown in Tab. 3. Across the 3 datasets, 60-70% of tasks have improved scores over the pretrained model. The distribution of test score improvements appear to be highly concentrated on a few tasks, with 20% of test tasks accounting for 60-80% of all improvement. The median score change for `UnpredicTable-5k` is only +2.8%, though the max is +43.0%.

Fig. 5 shows the 10 most-improving test tasks

[7]Random NLP tasks with $T = 40, N = 1250$ to match the total number of examples in `UnpredicTable-5k`.

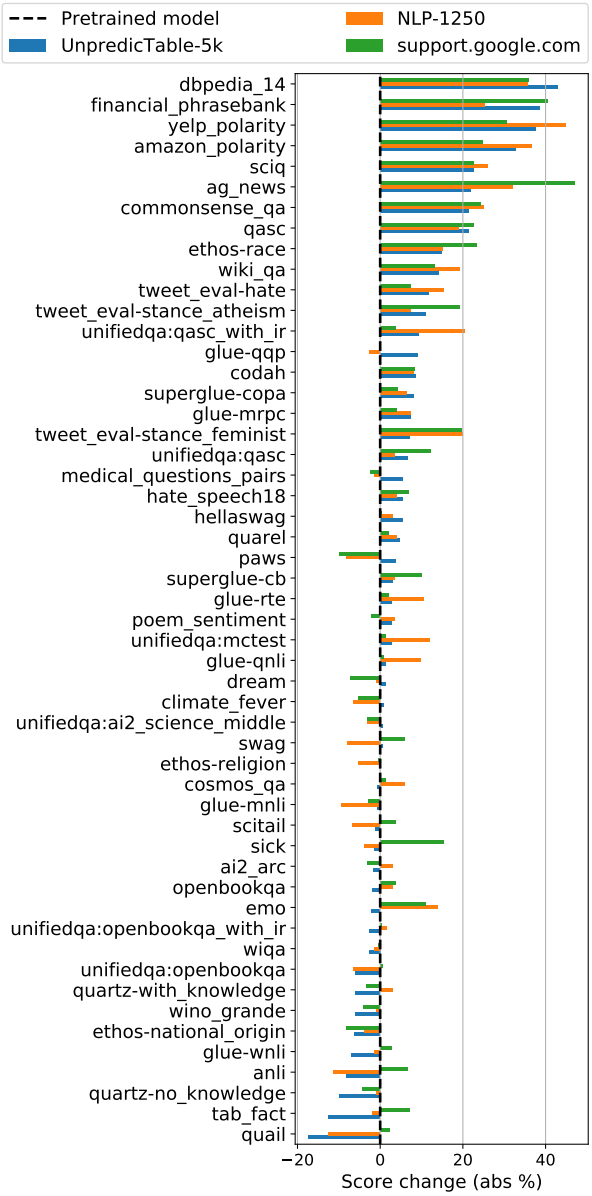


Figure 4: Score changes (vs pretrained) on 52 test tasks for models finetuned on 3 different datasets.

(median of all 90 training datasets in Fig. 4). The tasks are highly varied, spanning topics from finance to science, and have binary or multiple-choice (MCQ) labels. It is difficult to draw clear relationships between test tasks and the datasets that lead to their largest improvement **(Best dataset)**. For example, `cluster 7` (a dataset on web cookies) is the most helpful dataset for both `ag_news` (news classification) and `amazon_polarity` (sentiment classification). Our examples of unintuitive task transfer contradict prior work that suggest domain similarity is key for successful task transfer (Gururangan et al., 2020).

Task	Type	Output space	Chance (%)	Median (%)	Max (%)	Best dataset
ag_news	News class	World / Sports / Business / SciTech	25	42 (+29)	63 (+50)	cluster 7
dbpedia_14	Wikipedia class	14 classes (plant / athlete / ...)	7	31 (+25)	47 (+42)	w3.org
commonsense_qa	General QA	MCQ	20	44 (+23)	51 (+30)	cluster 12
sciq	Scientific QA	MCQ	25	81 (+23)	87 (+29)	cluster 0
amazon_polarity	Review class	positive / negative	50	77 (+18)	92 (+34)	cluster 7
qasc	General QA	MCQ	13	30 (+17)	38 (+25)	cluster 8
financial_phrasebank	Financial class	positive / negative / neutral	33	41 (+14)	68 (+40)	support.google.com
tweet_eval-stance_atheism	Tweet class	none / against / favor	33	31 (+13)	44 (+25)	msdn.microsoft.com
yelp_polarity	Review class	positive / negative	50	61 (+12)	84 (+36)	w3.org
ethos-race	Hate speech class	true / false	50	43 (+12)	55 (+23)	support.google.com

Figure 5: The most-improving tasks in the MetaICL test set span a wide variety of topics and output spaces. There is no clear connection to the training datasets that most strongly improve FSL performance (**Best dataset**), yet score improvements are significant. We show absolute scores for random **Chance** as well as the **Median** and **Max** scores across different training datasets. Improvements w.r.t. to the pretrained model are shown in parentheses.

	Table-5k	NLP-1250	support.google
	Test tasks counts (# out of 52)		
Improved	33	32	**37**
Decreased	19	20	**15**
>Chance (pre: 23)	23	31	**34**
	Score change (finetuned - pre) (%)		
Mean	+5.6	+6.7	**+7.5**
Median	+2.8	+3.5	**+3.6**
Max	+43.0	+44.7	**+47.1**
Min	-17.3	-12.5	**-10.0**

Table 3: *Top*: Rows 1 & 2 show the number of test tasks that improved or not (vs the pretrained model) after finetuning. Row 3 shows the number of test tasks that score >random chance for multiple-choice answers. *Bottom*: Improvements are not evenly distributed; the maximum score increase on `support.google.com` is +47.1% but median improvement is only +3.6%.

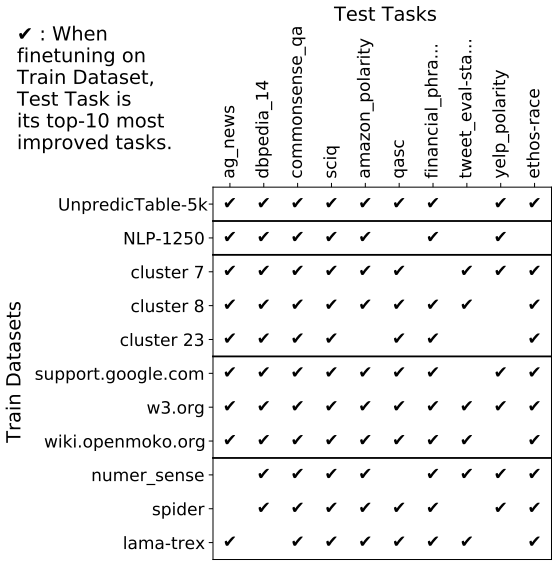


Figure 6: Finetuning on different datasets leads to broadly similar improvements. For example, finetuning on `wiki.openmoko.org` (software documentation) and `lama-trex` (factual knowledge) lead to 8 of the same test tasks being in their respective top-10 most-improved test tasks. (Out of 52 total test tasks)

4.6 Do different datasets lead improvements on different test tasks?

We wish to understand if finetuning on different datasets lead to different test task improvements. Fig. 6 illustrates that the same set of 10 test tasks make up the majority of the top-10 improving test tasks for each of our best training datasets (the top-performing datasets for each category in Fig. 4). This suggests that the improvements learned from highly different training datasets are domain-agnostic. However, it is unclear why these improvements can be learned from these particular training datasets but not others, and why these particular test tasks benefit most from the improvements.

5 Related Work

We focus on the FSL setting where few training samples are available. Pretrained LMs can learn from few-shot examples in-context (Brown et al., 2020; Scao and Rush, 2021) but have weaknesses including prompt sensitivity (Lu et al., 2021; Perez et al., 2021) and miscalibration (Zhao et al., 2021). Min et al. (2021) and Chen et al. (2021b) alleviate these issues with FSL adaptation - fine-tuning LMs to predict the target given few-shot examples in the prompt. We adopt MetaICL (Min et al., 2021) training for our main experiments and support our results with additional few-shot benchmarks, CrossFit (Ye et al., 2021) and FLEX (Bragg et al., 2021).

Our work connects with other work in domain adaptation. Gururangan et al. (2020) show that finetuning on domains related to the downstream task leads to performance gains. More recent examples include Chen et al. (2021a) for coding tasks and

Lewkowycz et al. (2022) for mathematics tasks. Solaiman and Dennison (2021) demonstrate finetuning on value-aligned text to generate text in accordance with intrinsic human values. In contrast, we show that LMs can be finetuned on unrelated domains to improve on new tasks. Other work adapt to task formats: Khashabi et al. (2020); Huber et al. (2021); Zhong et al. (2021b) convert broad NLP tasks into question-answering tasks and finetune to excel at question-answering; Zhong et al. (2021a) finetune models for classification tasks; Gao et al. (2020) finetune models to perform tasks within predetermined prompt templates. More generally, LMs have been finetuned to follow instructions (Ouyang et al., 2022; Wei et al., 2021) which allows for diverse task formats. FSL adaptation can be seen as adaptation to the FSL prompt format, though the tasks can be diverse in domain and structure.

Multi-task literature show that training on a wide variety of tasks improves generalization to new tasks, which motivates our exploration of a large scale task dataset. Sanh et al. (2021); Aribandi et al. (2021); Mishra et al. (2021); Aghajanyan et al. (2021a); Padmakumar et al. (2022) demonstrate that increasing the number of tasks for multi-task training improves generalization in the zero-shot setting. Xu et al. (2022); Wang et al. (2022) extended this result to more than 1,000 tasks. We were inspired by these results to obtain a training dataset with 100x more tasks, but found certain narrow datasets are more helpful than diverse ones. Padmakumar et al. (2022) showed that some training tasks negatively impact downstream performance, which could explain why mixing diverse tasks might underperform. This begs the question of how to select training datasets to improve downstream task performance. Vu et al. (2020) show that domain similarity can be used as a predictor for successful transfer, but our results suggest there may be domain-agnostic improvements to be gained from training on tasks unrelated to the test tasks. Others study the effect of pretraining data on FSL, including (Shin et al., 2022) and (Chan et al., 2022) who find that FSL emerges when the training data exhibits particular distributional properties.

Our use of structured datasets to generate training tasks is inspired by other work, though others have focused on a limited set of task types. Yoran et al. (2021) also turn tables into tasks, using handwritten templates to extract question-answer pairs from tables. Aghajanyan et al. (2021b) train LMs to predict masked spans in HTML webpages, then use HTML markup to prompt language models to do summarization and classification tasks. Chen et al. (2022) transform ordinary (non-table) text into sentence completion, masked phrase prediction, and classification tasks. In contrast, our approach captures any tasks that occur naturally in tables.

6 Limitations & Future Work

The `UnpredicTable` dataset may contain inaccuracies, biases, and inappropriate content. We do not recommend using this dataset to train models for deployment, but release this primarily as a research resource. We do not introduce any new model capabilities that lead to different risks than the usual risks associated with model usage. Our work highlights the unpredictability of model behavior given various training datasets which calls for heightened vigilance for behavior changes after finetuning. Our design choices in using table data for FSL training led to a dataset that is quite different than typical NLP datasets, so specific results from training on our dataset may not fully generalize to other kinds of datasets. Further work may consider other methods for converting tables to tasks, other sources of tables besides WTC, or other structured datasets besides tables. Our experiments focused on modestly-sized models (GPT-2 Large, 750M parameters) so our conclusions may not hold for larger models. Our evaluations are limited to multiple-choice tasks. Future work may extend our analyses with larger models and other tasks including freeform generation.

7 Conclusion

We produced `UnpredicTable`, a dataset of 413,299 diverse few-shot learning tasks from internet tables. Finetuning on `UnpredicTable` improves the FSL ability of LMs. However, the size of our dataset is not the key factor in its success. We find that certain narrow datasets (even ones made of trivia) are even more helpful than diverse, curated NLP datasets. Finetuning on these narrow datasets leads to strong improvements on the same test tasks as finetuning on diverse, curated NLP datasets. This suggests that finetuning on these datasets cause domain-agnostic FSL gains, though we were unable to find clear patterns to explain why this happens for some data and not others. Our results question common wisdom that task diversity is necessary for adapting LMs to FSL. We hope

our work spurs investigation on what data causes few-shot learning to emerge, both to develop better datasets and to better understand how training data leads to unexpected behaviors or failures.

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A Tables-to-tasks filtering

Below, we describe the filtering steps applied when converting tables into tasks:

Filtering tables We reject tables with fewer than 2 unique columns (one for the task output and at least one more for the input) or 6 unique rows (at least 5 examples + 1 target row). We find a large number of tables containing junk data or only numerical values. To remove these, we reject tables with $\geq 20\%$ of tokens tagged as either *Numeral*, *Proper Noun*, *Symbol*, *Punctuation*, or *Other* by the `spaCy` part-of-speech classifier.[8] The tables that pass this filtering stage are converted into tasks.

Filtering tasks Given a set of candidate tasks, we require that the output space contains at least two unique answers, and reject tasks with severe class imbalance.[9] To narrow our scope to tasks with a single correct answer, we reject tasks where any input appears more than once with different outputs. Finally, we only accept up to 2500 tasks per website to counter imbalance[10] in the source website of generated tasks. Appendix A shows the breakdown of items filtered at each stage.

Tab. 4 shows the number of tables and tasks filtered at each stage of our tables-to-tasks procedure.

tables initial	50,820,216
rejected min rows	−25,638,244
rejected non-english	−23,034,542
tables remaining	2,147,532
tasks initial	5,646,614
rejected max domain	−4,054,764
rejected min rows	−99,226
rejected one-to-many	−322,536
rejected min classes	−157,199
rejected non-english output	−561,622
rejected class balance	−38,505
tasks remaining	413,299

Table 4: Converting 50M tables into 400k tasks.

B Dataset License

The WDC Web Table Corpus 2015 dataset is provided under the Apache-2.0 license. Our usage of the dataset is in accordance with intended use which includes NLP research (Lehmberg et al., 2016). Our dataset, UnpredicTable, is likewise released with the Apache-2.0 license.

C MetaICL experiment details

This section provides training and evaluation details for our MetaICL experiments in §3 and §4. The datasets used in MetaICL train and test settings are taken from CROSSFIT (Ye et al., 2021) and UNIFIEDQA (Khashabi et al., 2020), which in turn have been compiled from various other sources. The full list for all datasets and their citations are provided in Fig. 7. We make use of 3 different task splits:

Test Tasks (52 tasks) The union of all test tasks from the 7 task settings in Min et al. (2021).

Train Tasks (90 tasks) Contains all tasks in Min et al. (2021) except those which are Test Tasks. These tasks are only used as a source of NLP datasets in §4.

Dev Tasks (50 tasks) Contains all our Train Tasks except those which are not multiple-choice. These tasks are used for hyperparameter selection.

For hyperparameter selection, we fine-tune the GPT2-large model (774M)[11] on `UnpredicTable-5k` and sweep over batch sizes $\{1, 8, 64\}$ and learning rates $\{5e^{-5}, 5e^{-6}, 5e^{-7}\}$. We select batch size = 1 and learning rate = $5e^{-6}$ based on Dev scores and use this for all MetaICL experiments. We train for 5 epochs and evaluate after each epoch, selecting the checkpoint with the highest mean Dev Tasks score. We report scores of the selected checkpoint evaluated on the Test Tasks. Each training and inference run is done on a single RTX8000 GPU. The duration of training varies by dataset size (training 5 epochs on `UnpredicTable-5k` takes ∼24 hours).

D Do Other Learning Algorithms Benefit from Table Data?

Our main experiments use the MetaICL algorithm and benchmarks for training and evaluation. To understand how well our findings hold in other settings, we report additional experiments comparing `UnpredicTable-5k` against NLP datasets using different multi-task learning algorithms, models, and evaluation settings.

[8] spacy.io/usage/linguistic-features#pos-tagging

[9] We reject classes with Shannon Diversity Index ≤0.7.

[10] Without rebalancing, 41% of tasks are from cappex.com.

[11] GPT2-large LM https://huggingface.co/gpt2-large

D.1 MetaICL zero-shot

We investigate whether finetuning on our dataset also helps in the zero-shot generalization case. We use a similar setup as §4 where $\mathcal{D}_{\text{test}}$ contains all 52 test tasks from the MetaICL test set and we compare between $\mathcal{D}_{\text{train}}$ of `UnpredicTable-5k`, `NLP-1250` and `support.google.com`. Instead of few-shot (**FS**) as before, we now use the models zero-shot (**ZS**) i.e. $k = 0$ so the model is trained to maximize $\log P(y_i|x_i)$ for each training pair (x_i, y_i). At test time, the model selects the most likely label y for an unseen query x.

$\mathcal{D}_{\text{train}}$	**ZS**	**FS**
Pretrained (GPT2-large)	34.5	35.6
`NLP-1250`	39.1	42.3
`UnpredicTable-5k`	38.7	40.6
`support.google.com`	39.7	43.1

Table 5: Comparing zero-shot (ZS) and few-shot (FS) methods for the pretrained model, finetuning on NLP datasets (`NLP-1250`), and finetuning on table datasets (`UnpredicTable-5k`, `support.google.com`). Showing mean scores on 52 test tasks.

Results Tab. 5 compares fine-tuning on 3 different datasets using two methods: **ZS** and **FS** (FS results same as Tab. 3). Scores are the mean over 52 test tasks. We find that finetuning on our table datasets (`UnpredicTable-5k` and `support.google.com`) is as effective as finetuning on NLP datasets (`NLP-1250`) for improving zero-shot generalization. Notably, as in the few-shot case, training on `support.google.com` improves zero-shot performance (+5.2%) even more than training on curated NLP datasets (`NLP-1250`) (+4.6%). This result validates that the benefit of training on our table datasets is not a quirk of our particular FSL training setup, but also applies to the more general zero-shot setting.

D.2 CrossFit

Ye et al. (2021) introduce the Few-Shot Gym, a collection of 160 NLP tasks, and a problem setup called CrossFit. We focus on the *Random* task partition of CrossFit where $\mathcal{D}_{\text{train}}$ and $\mathcal{D}_{\text{test}}$ contain 120 and 20 tasks respectively, sampled IID from the Few-Shot Gym. For our learning algorithm, we adopt the best-performing method in Ye et al. (2021), MTL, which finetunes on $\mathcal{D}_{\text{train}}$ followed by finetuning on the few-shot training examples from a given target task in $\mathcal{D}_{\text{test}}$ (finetuning a separate model for each target task in $\mathcal{D}_{\text{test}}$). We compare three different methods: MTL with $\mathcal{D}_{\text{train}}$ from the Few-Shot Gym, MTL with `UnpredicTable-5k` as $\mathcal{D}_{\text{train}}$, and Direct Finetuning (DF) which is a baseline without finetuning on any $\mathcal{D}_{\text{train}}$. All experiments finetune a BART-Base (Lewis et al., 2019), a pretrained encoder-decoder transformer model (Vaswani et al., 2017).

Task	**DF**	**MTL**	**Ours**
glue-cola	0.0	1.0	0.0
crawl_domain	30.6	25.6	29.5
ag_news	86.1	82.6	84.9
ai2_arc	16.1	25.4	15.7
wiki_split	79.6	80.0	78.4
amazon_polarity	79.4	92.1	90.8
blimp-..._present	99.4	98.5	97.8
tweet_eval-irony	55.0	56.4	52.5
ethos-disability	75.8	77.7	71.3
sglue-rte	49.5	56.2	49.9
circa	46.3	44.8	48.3
ethos-sexual_orient.	57.7	69.9	60.9
hatexplain	42.0	45.5	41.0
race-high	16.5	32.4	14.2
glue-qnli	60.5	74.2	56.9
quoref	24.7	41.8	23.3
blimp-...npi_scope	70.9	97.1	82.6
break-QDMR	2.3	4.8	1.7
yelp_polarity	40.6	93.5	56.2
freebase-qa	0.5	1.2	0.4
mean	**46.7**	**49.1**	**47.8**

Table 6: Results on the CrossFit benchmark. We compare the Direct Finetuning **DF** baseline (no multi-task learning) against multi-task learning on the NLP Few-shot Gym dataset (**MTL**) and multi-task learning with `UnpredicTable-5k` (**Ours**).

Results Tab. 6 shows the full results. Compared to DF, MTL with our dataset improves results by a mean of +1.1%. 3 out of 20 tasks improve by more than +10% including `amazon_polarity` and `yelp_polarity`, which are also among the tasks with the largest improvements in MetaICL. MTL with `UnpredicTable-5k` is less helpful than MTL with curated NLP datasets (+2.4% relative to DF), but still recovers 46% of the relative improvement from finetuning on 120 curated NLP tasks. Our results show that finetuning on UnpredicTable helps even with MTL (a different learning algorithm) on BART (a different LM). We see large

gains on similar tasks as in MetaICL, which suggests that our data helps consistently on these tasks (and the observed gains are not just an artifact of MetaICL training).

D.3 FLEX

FLEX (Bragg et al., 2021) is a FSL benchmark that provides 11 NLP training tasks and 20 NLP test tasks, carefully chosen to evaluate various task transfer settings. The baseline model is **UniFew**, which uses a UnifiedQA model (Khashabi et al., 2020) with a prompt that converts task examples into a multiple-choice question-answer format. The primary FLEX model is **UniFew$_{\text{Meta}}$**, which is UniFew finetuned with the 11 FLEX training tasks. As in MetaICL, UniFew$_{\text{Meta}}$ finetuning uses k examples in the input to maximize $\log P(y_{k+1}|x_1, y_1, \ldots, x_k, y_k, x_{k+1})$. Our approach (**Ours**) uses the same setup as UniFew$_{\text{Meta}}$ but replaces the FLEX training tasks with `UnpredicTable-5k`. Evaluation for all models is done with FSL on the FLEX test tasks.

Task	UniFew	Ours	UniFew$_{\text{Meta}}$
FewRel	79.2	79.4	87.2
HuffPost	62.8	63.1	68.0
Amazon	79.5	79.4	82.1
20News	63.1	63.4	67.3
Reuters	94.5	95.5	96.3
MR	78.6	83.1	89.4
CR	90.1	92.0	93.3
SNLI	55.8	56.5	80.9
SciTail	64.9	65.5	83.6
SUBJ	60.5	63.7	68.7
TREC	58.1	62.9	60.0
CoNLL	44.3	44.0	58.6
Mean	**69.3**	**70.7**	**77.9**

Table 7: Results on the FLEX benchmark. We compare the pretraining-only **UniFew** model against the same model finetuned on the FLEX dataset (**Unifew-Meta**) and `UnpredicTable-5k` (**Ours**).

Results Tab. 7 shows our results. Training on our dataset improves over UniFew for 10/12 tasks (mean +1.4%, max +5.5%). However, we do not approach the level of UniFew$_{\text{Meta}}$ (mean improvement +8.6%). This discrepancy is likely because the FLEX training and test tasks have been chosen with overlapping domains/task types to study various transfer learning settings (see Bragg et al. (2021) for details). Nevertheless, the results show that our table tasks still lead to improvements in FLEX with a different model and test tasks.

E Clustering

Here, we describe the clustering procedure used to group `UnpredicTable-unique` tasks into narrow data subsets based on content. For all examples in all tasks, we concatenate each (x, y) example and obtain their embeddings from a pretrained GPT-2 model[12]. We average the resulting 1024-dimensional embeddings at a task level. We normalize each task embedding and apply a two-stage dimensionality reduction consisting of a PCA transformation to 128 dimensions followed by further reduction using UMAP (McInnes et al. (2018), $n_{\text{neighbors}} = 4$, $\text{d}_{\text{min}} = 0.0$) to 32 dimensions. We cluster the 32D task embeddings using the HDBSCAN algorithm (McInnes et al., 2017) with a minimum cluster size of 60 and 400 minimum samples. This setup results in 30 task clusters plus an additional cluster (`cluster -1`) containing tasks that HDBSCAN rejected as noise. The cluster sizes range from $T = 61$ to $T = 5700$. We tested several hyperparameters for our clustering pipeline until we arrived at a setup with reasonable in-cluster content similarity (manual inspection).

F Task Quality Annotation Instructions

Below, we display a condensed version of the instructions given to annotators for annotating the dataset into different task quality levels. The full instructions are available online[13].

Introduction Thank you for agreeing to contribute annotations to our dataset! Here are some brief instructions to help you successfully complete this work.

Context We have a large number of **Tasks** created for training language models to learn a variety of skills. A standard example of a task is shown in Tab. 8 as Task 1. This example closely resembles the Question-Answer form that is commonly encountered in human competency tests, but this is not the only valid form. More generally, a **Task** is simply a set of **input-output** pairs where the inputs map to outputs in a common and (given knowledge

[12] `stanford-crfm/eowyn-gpt2-medium-x777` via the HuggingFace Transformers library.

[13] Full instructions for task quality annotations: `https://bit.ly/3veIWf7`

of the mapping) predictable way; given an input, an individual skilled in this task should be able to respond with the correct output. Another example of a valid task is shown in Tab. 8 as Task 2. In this case, the inputs are a set of issues that a user might be having, and the outputs suggest actions to address each issue.

Examples of Tasks for Annotation	
Task 1	
input	[Question] The parotid glands are located: [Answer]
output	cheek
input	[Question] The roof of the mouth is called the: [Answer]
output	hard palte
input	[Question] The bone that forms the posterior portion of the skull is the [Answer]
output	occipital bone
input	[Question] The lower jawbone is the [Answer]
output	mandible
Task 2	
input	[If you want to ...] Get a page or site removed from Google [Then ...]
output	Submit a URL removal request.
input	[If you want to ...] Report spam [Then ...]
output	Submit a spam report.
input	[If you want to ...] Report a copyright violation or the misuse of your content [Then ...]
output	File a DMCA takedown request.
input	[If you want to ...] Tell Google to crawl your site more slowly [Then ...]
output	Request a change in crawl rate.
input	[If you want to ...] Tell Google that your content is mistakenly being filtered by SafeSearch [Then ...]
output	Submit a SafeSearch issue.

Table 8: Example tasks provided with the instructions for the task-quality annotation

The Problem Our pool of tasks has been curated in an automated way from natural internet content, so they vary greatly in quality and form. It would be valuable to label each task's quality so that we may investigate (1) what is the overall quality in our pool of tasks, and (2) how task quality affects the ability of language models to learn from it.

The Work In this session, you will classify a number of tasks in terms of how feasible and useful they are. Each task should be rated from 0-2, where 0 is "This task is not valid or useful at all" and 2 is "This task demonstrates an interesting and useful skill".

Criteria of Class 0 (low rating) Tasks

- The input-output mapping appears nonsensical and/or arbitrary.
- The task is not in English.
- Would never be useful in any realistic setting / practicing this task does not build any generally-useful skills.
- Tests highly obscure knowledge that is not correlated with the input text (highly context-dependent knowledge, entertainment trivia on fan sites, product specifications, . . .)
- You would not even be able to tell if all output labels have been shuffled.

Criteria of Class 1 (medium rating) Tasks

- This class is a catch-all for tasks that are neither squarely Class 0 nor Class 2.
- The task is quite interesting, but its current form contains flaws that make it confusing or lacks enough context to do a good job of the task.
- You could narrow the space of possible options and guess the right answer with better-than-random accuracy (especially with the help of multiple-choice options).
- The task makes sense but is trivial or not interesting enough to be Class 2. For example, the output is just a copy of the input.

Criteria of Class 2 (high rating) Tasks

- The task is well-posed with enough context that an expert could give a reasonably correct answer most of the time.

- Demonstrates a skill that is definitely useful for real-world tasks, i.e. might be tested in an exam or competency test, or part of a job.
- Resembles the type of skill that is tested in typical NLP datasets. See "Examples from real NLP datasets" section in the full instructions[13].

Further notes

- These criteria are not a complete set of rules for membership, so based on the above you may make your own judgement regarding a new task that does not perfectly fit any criteria.
- We expect that the majority of our tasks will fall into either Class 0 or Class 1; fewer than 20% of the tasks will meet the standard for Class 2.
- A single input may not always be enough to know what the task expects in the output; this is acceptable (even for Class 2) as long as the input-output mapping is clear after observing several demonstration pairs.
- The "Examples from real NLP datasets" section in the full instructions[13] show the kinds of interesting tasks we would like to see in Class 2, but we expect (and encourage) that our tasks will span a wider variety that are still interesting and valuable.

G Examples of tasks

In the following pages, we provide examples from various datasets discussed in the text:

1. Quality-annotated (High)
2. Quality-annotated (Med)
3. Quality-annotated (Low)
4. Single-website (support.google.com)
5. Single-website (w3.org)
6. Single-website (mmo-champion)
7. Single-website (studystack.com)
8. Cluster 7
9. Cluster 8
10. Cluster -1
11. Cluster 3
12. NLP train (2 best and 2 worst)
13. NLP test (10 most-improving)

Train Tasks (90 tasks)

ade_corpus_v2-classification (Gurulingappa et al., 2012), ade_corpus_v2-dosage (Gurulingappa et al., 2012), art (Bhagavatula et al., 2020), biomrc (Pappas et al., 2020), blimp-anaphor_number_agreement (Warstadt et al., 2020), blimp-ellipsis_n_bar_2 (Warstadt et al., 2020), blimp-sentential_negation_npi_licensor_present (Warstadt et al., 2020), blimp-sentential_negation_npi_scope (Warstadt et al., 2020), boolq (Clark et al., 2019), circa (Louis et al., 2020), crows_pairs (Nangia et al., 2020), discovery (Sileo et al., 2019), emotion (Saravia et al., 2018), ethos-directed_vs_generalized (Mollas et al., 2020), ethos-disability (Mollas et al., 2020), ethos-gender (Mollas et al., 2020), ethos-sexual_orientation (Mollas et al., 2020), freebase_qa (Jiang et al., 2019), gigaword (Napoles et al., 2012), glue-cola (Warstadt et al., 2019), glue-sst2 (Socher et al., 2013), google_wellformed_query (Faruqui and Das, 2018), hate_speech_offensive (Davidson et al., 2017), hatexplain (Mathew et al., 2020), health_fact (Kotonya and Toni, 2020), hotpot_qa (Yang et al., 2018), imdb (Maas et al., 2011), kilt_ay2 (Hoffart et al., 2011), kilt_fever (Thorne et al., 2018), kilt_hotpotqa (Yang et al., 2018), kilt_nq (Kwiatkowski et al., 2019), kilt_trex (Elsahar et al., 2018), kilt_zsre (Levy et al., 2017), lama-conceptnet (Petroni et al., 2019, 2020), lama-google_re (Petroni et al., 2019, 2020), lama-squad (Petroni et al., 2019, 2020), lama-trex (Petroni et al., 2019, 2020), liar (Wang, 2017), mc_taco (Zhou et al., 2019), numer_sense (Lin et al., 2020), onestop_english (Vajjala and Lučić, 2018), piqa (Bisk et al., 2020), proto_qa (Boratko et al., 2020), qa_srl (He et al., 2015), quoref (Dasigi et al., 2019)12, race-high (Lai et al., 2017), race-middle (Lai et al., 2017), ropes (Lin et al., 2019), rotten_tomatoes (Pang and Lee, 2005), search_qa (Dunn et al., 2017), sms_spam (Almeida et al., 2011), social_i_qa (Sap et al., 2019a), spider (Yu et al., 2018), squad-no_context (Rajpurkar et al., 2016), squad-with_context (Rajpurkar et al., 2016), superglue-multirc (Khashabi et al., 2018), superglue-record (Zhang et al., 2018), superglue-rte (Dagan et al., 2005; Bar-Haim et al., 2006)(Giampiccolo et al., 2007; Bentivogli et al., 2009), superglue-wic (Pilehvar and Camacho-Collados, 2019), superglue-wsc (Levesque et al., 2012), trec (Li and Roth, 2002; Hovy et al., 2001), trec-finegrained (Li and Roth, 2002; Hovy et al., 2001), tweet_eval-emoji (Barbieri et al., 2020), tweet_eval-emotion (Barbieri et al., 2020), tweet_eval-irony (Barbieri et al., 2020), tweet_eval-offensive (Barbieri et al., 2020), tweet_eval-sentiment (Barbieri et al., 2020), tweet_eval-stance_abortion (Barbieri et al., 2020), tweet_eval-stance_climate (Barbieri et al., 2020), tweet_eval-stance_hillary (Barbieri et al., 2020), tweet_qa (Xiong et al., 2019), unifiedqa:boolq (Clark et al., 2019), unifiedqa:commonsenseqa (Talmor et al., 2019), unifiedqa:drop (Dua et al., 2019), unifiedqa:narrativeqa (Kociský et al., 2018), unifiedqa:natural_questions_with_dpr_para, unifiedqa:newsqa (Trischler et al., 2017), unifiedqa:physical_iqa (Bisk et al., 2020), unifiedqa:quoref (Dasigi et al., 2019), unifiedqa:race_string (Lai et al., 2017), unifiedqa:ropes (Lin et al., 2019), unifiedqa:social_iqa (Sap et al., 2019b), unifiedqa:squad1_1 (Rajpurkar et al., 2016), unifiedqa:squad2 (Rajpurkar et al., 2018), unifiedqa:winogrande_xl (Sakaguchi et al., 2020a), web_questions (Berant et al., 2013), wikisql (Zhong et al., 2017), xsum (Narayan et al., 2018), yahoo_answers_topics (link), yelp_review_full (Zhang et al., 2015)

Test Tasks (52 tasks)

ag_news Gulli (link), ai2_arc (Clark et al., 2018), amazon_polarity (McAuley and Leskovec, 2013), anli (Nie et al., 2020), climate_fever (Diggelmann et al., 2020), codah (Chen et al., 2019), commonsense_qa (Talmor et al., 2019), cosmos_qa (Huang et al., 2019), dbpedia_14 (Lehmann et al., 2015), dream (Sun et al., 2019), emo (Chatterjee et al., 2019), ethos-national_origin (Mollas et al., 2020), ethos-race (Mollas et al., 2020), ethos-religion (Mollas et al., 2020), financial_phrasebank (Malo et al., 2014), glue-mnli (Williams et al., 2018), glue-mrpc (Dolan and Brockett, 2005), glue-qnli (Rajpurkar et al., 2016), glue-qqp (`data.quora.com/First-Quora-Dataset-Release-Question-Pairs`), glue-rte (Dagan et al., 2005; Bar-Haim et al., 2006)(Giampiccolo et al., 2007; Bentivogli et al., 2009), glue-wnli (Levesque et al., 2012), hate_speech18 (de Gibert et al., 2018), hellaswag (Zellers et al., 2019), medical_questions_pairs (McCreery et al., 2020), openbookqa (Mihaylov et al., 2018), paws (Zhang et al., 2019), poem_sentiment (Sheng and Uthus, 2020), qasc (Khot et al., 2020), quail (Rogers et al., 2020), quarel (Tafjord et al., 2019a), quartz-no_knowledge (Tafjord et al., 2019b), quartz-with_knowledge (Tafjord et al., 2019b), sciq (Welbl et al., 2017), scitail (Khot et al., 2018), sick (Marelli et al., 2014), superglue-cb (de Marneffe et al., 2019), superglue-copa (Gordon et al., 2012), swag (Zellers et al., 2018), tab_fact (Chen et al., 2020), tweet_eval-hate (Barbieri et al., 2020), tweet_eval-stance_atheism (Barbieri et al., 2020), tweet_eval-stance_feminist (Barbieri et al., 2020), unifiedqa:ai2_science_middle (`data.allenai.org/ai2-science-questions`), unifiedqa:mctest (Richardson et al., 2013), unifiedqa:openbookqa (Mihaylov et al., 2018), unifiedqa:openbookqa_with_ir, unifiedqa:qasc (Khot et al., 2019), unifiedqa:qasc_with_ir, wiki_qa (Yang et al., 2015), wino_grande (Sakaguchi et al., 2020b), wiqa (Tandon et al., 2019), yelp_polarity (Zhang et al., 2015)

Dev Tasks (50 tasks)

ade_corpus_v2-classification, art, biomrc, blimp-anaphor_number_agreement, blimp-ellipsis_n_bar_2, blimp-sentential_negation_npi_licensor_present, blimp-sentential_negation_npi_scope, boolq, circa, crows_pairs, discovery, emotion, ethos-directed_vs_generalized, ethos-disability, ethos-gender, ethos-sexual_orientation, glue-cola, glue-sst2, google_wellformed_query, hate_speech_offensive, hatexplain, health_fact, imdb, kilt_fever, liar, mc_taco, numer_sense, onestop_english, piqa, race-high, race-middle, rotten_tomatoes, sms_spam, social_i_qa, superglue-multirc, superglue-rte, superglue-wic, superglue-wsc, trec, trec-finegrained, tweet_eval-emoji, tweet_eval-emotion, tweet_eval-irony, tweet_eval-offensive, tweet_eval-sentiment, tweet_eval-stance_abortion, tweet_eval-stance_climate, tweet_eval-stance_hillary, yahoo_answers_topics, yelp_review_full

Figure 7: All the task datasets used in our MetaICL experiments, along with citations of their original source. Dev Tasks are a subset of Train Tasks so citations are not repeated.

quality_annotated : High	
Task 1 (6 examples)	
input	[Format option] Heading 3 [What it will look like]
output	is a sub-header and can be used as a sub-section heading
input	[Format option] Code / preformatted [What it will look like]
output	Technical text that should be displayed in a fixed-width font
input	[Format option] Heading 5 [What it will look like]
output	is the smallest sub-header option
Task 2 (10 examples)	
input	[No.] 07 [Answer] Sahara desert [Question]
output	The biggest desert in the world is the
input	[No.] 02 [Answer] Nile [Question]
output	The longest river in the world is the
input	[No.] 05 [Answer] Everest [Question]
output	The highest mountain in the world is the
Task 3 (6 examples)	
input	[property] monitorType [applies to] all [description] one of counter, guage, string [type]
output	enum
input	[property] observedAttribute [applies to] all [description] the attribute being observed [type]
output	string
input	[property] initThreshold [applies to] counter [description] initial threshold value [type]
output	number
Task 4 (14 examples)	
input	[Verse] 14 [King James Version] And she lay at his feet until the morning: and she rose up before one could know another. And he said, Let it not be known that a woman came into the floor. So she lay at his feet until morning. She got up before either could know the other. He said, "Don't let it be known that a woman came into the threshing-floor." [Analysis]
output	Boaz wants to avoid scandal.
input	[Verse] 5 [King James Version] And she said unto her, All that thou sayest unto me I will do. Ruth said to her, "I will do everything you say." [Analysis]
output	What Ruth must have thought of these orders, none can speculate.
input	[Verse] 1 [King James Version] Then Naomi her mother in law said unto her, My daughter, shall I not seek rest for thee, that it may be well with thee? Now Naomi, mother-in-law of Ruth, said to her, "My daughter, I should find you a place of rest, that will be good for you. [Analysis]
output	Naomi wants to settle Ruth properly.

quality_annotated : Med	
Task 1 (11 examples)	
input	[Symptom] Sore Throat [Cold] Sore throat is commonly present with a cold. [Flu] Sore throat is not commonly present with the flu. [Allergies]
output	Sore throat is sometimes present if enough post-nasal drainage occurs.
input	[Symptom] Sudden Symptoms [Cold] Cold symptoms tend to develop over a few days. [Flu] The flu has a rapid onset within 3-6 hours. The flu hits hard and includes sudden symptoms like high fever, aches and pains. [Allergies]
output	Rapid onset.
input	[Symptom] Aches [Cold] Slight body aches and pains can be part of a cold. [Flu] Severe aches and pains are common with the flu. [Allergies]
output	No aches and pains.
Task 2 (9 examples)	
input	[0] Space Requirements Larger due to the existence of aggregation structures and history data; requires more indexes than OLTP
output	Can be relatively small if historical data is archived
input	[0] Backup and Recovery Instead of regular backups, some environments may consider simply reloading the OLTP data as a recovery method
output	Backup religiously; operational data is critical to run the business, data loss is likely to entail significant monetary loss and legal liability
input	[0] Queries Often complex queries involving aggregations
output	Relatively standardized and simple queries Returning relatively few records
Task 3 (7 examples)	
input	[Action] Add a point to an editable shape [Shortcut]
output	Option-click the shape edge where you want to add a point
input	[Action] Change a curved point of an editable shape into a corner point [Shortcut]
output	Double-click the curved point
input	[Action] Delete a point of an editable shape [Shortcut]
output	Click point and press Delete
Task 4 (8 examples)	
input	[0] Length [1] meter [2]
output	distance light travels in a vacuum
input	[0] Time [1] second [2]
output	oscillations of the cesium atom
input	[0] Electric current [1] ampere [2]
output	attraction between two wires

quality_annotated : Low	
Task 1 (285 examples)	
input	[Career Cluster] Manufacturing [Career Title] Stationary Engineers and Boiler Operators [Nontraditional for...]
output	Women
input	[Career Cluster] Health Science [Career Title] Health Care Social Workers [Nontraditional for...]
output	Men
input	[Career Cluster] Government and Public Administration [Career Title] Government Program Eligibility Interviewers [Nontraditional for...]
output	Men
Task 2 (8 examples)	
input	[RESTRICTED] YES CONFIDENTIAL [UNRESTRICTED]
output	NO (Sensitive/need to know)
input	[RESTRICTED] Available COUNSELING SERVICES [UNRESTRICTED]
output	Available
input	[RESTRICTED] Active Duty Military Only ELIGIBILITY [UNRESTRICTED]
output	All personnel
Task 3 (6 examples)	
input	[Talent Cards] Beat Back [Type]
output	Melee
input	[Type]
output	Insanity
input	[Talent Cards] Clear Minded [Type]
output	Focus
Task 4 (10 examples)	
input	[Directive] odbc.default_db [Master Value] no value [Local Value]
output	no value
input	[Directive] odbc.defaultlrl [Master Value] return up to 4096 bytes [Local Value]
output	return up to 4096 bytes
input	[Directive] odbc.defaultbinmode [Master Value] return as is [Local Value]
output	return as is

single_website_tables : support.google.com	
Task 1 (6 examples)	
input	[If you want to ...] Report a copyright violation or the misuse of your content [Then ...]
output	File a DMCA takedown request.
input	[If you want to ...] Tell Google to crawl your site more slowly [Then ...]
output	Request a change in crawl rate.
input	[If you want to ...] Get a site added back to Google [Then ...]
output	If your site was distributing malware, and is now clean, request a malware review. If your site was showing spam, but is now clean, submit a reconsideration request. If your site was in violation of the Webmaster Guidelines, but is now clean, submit ... *(Truncated)*
Task 2 (6 examples)	
input	[Term] Impressions [Search Console usage] Used exclusively for Google Search impressions [Analytics usage]
output	Used for both AdWords impressions and Google Search impressions
input	[Term] CTR [Search Console usage] Clickthrough rate. Clicks/Impressions for Google Search clicks. [Analytics usage]
output	Clickthrough rate. Clicks/Impressions for both AdWords and Google Search clicks.
input	[Term] Average Position [Search Console usage] Average ranking in Google Search results [Analytics usage]
output	Average ranking in Google Search results
Task 3 (7 examples)	
input	[Setting] Devices [Description] Campaigns target all types of devices, which include desktops, tablets, and mobile devices. Later, you can choose to customize ads for different devices. [Learn more]
output	Types of mobile ads
input	[Setting] Locations and languages [Description] Your campaign's ads are eligible to show to customers in your targeted geographic locations, or to customers who have selected your targeted language as their interface language. We recommend choosing t ... *(Truncated)*
output	Location and language targeting
input	[Setting] Type [Description] The campaign type determines which settings we'll show you as you create or edit your campaign. The type you choose tailors the campaign setup to just what's appropriate for your goals, eliminating unrelated features. We ... *(Truncated)*
output	Choosing the campaign type that's right for you
Task 4 (6 examples)	
input	[Then ...] File a DMCA takedown request. [If you want to ...]
output	Report a copyright violation or the misuse of your content
input	[Then ...] Submit a URL removal request. [If you want to ...]
output	Get a page or site removed from Google
input	[Then ...] If your site was distributing malware, and is now clean, request a malware review. If your site was showing spam, but is now clean, submit a reconsideration request. If your site was in violation of the Webmaster Guidelines, but is now cle ... *(Truncated)*
output	Get a site added back to Google

single_website_tables : w3.org	
Task 1 (23 examples)	
input	[Keyword] week [Data type] A date consisting of a week-year number and a week number with no time zone [Control type] A week control [State]
output	Week
input	[Keyword] hidden [Data type] An arbitrary string [Control type] n/a [State]
output	Hidden
input	[Keyword] password [Data type] Text with no line breaks (sensitive information) [Control type] A text field that obscures data entry [State]
output	Password
Task 2 (6 examples)	
input	[Attribute Name] next [Details]
output	an ECMAScript expression which returns the URI of the CCXML document to be fetched.
input	[Attribute Name] timeout [Details]
output	is an ECMAScript expression returning a string in CSS2 [CSS2] format interpreted as a time interval. The interval begins when the is executed. The fetch will fail if not completed at the end of this interval. A failed fetch will return the error.fetc ... *(Truncated)*
input	[Attribute Name] synch [Details]
output	is an ECMAScript left-hand-side expression that is set to the fetch completion event. The specification of this attribute in a implies a blocking fetch, which will be executed synchronously. If this attribute is not specified, the fetch is asynchrono ... *(Truncated)*
Task 3 (7 examples)	
input	[Function] DeleteScope [Arguments] name(optional) [Description] Removes a scope from the scope stack. If no name is provided, the topmost scope is removed. Otherwise the scope with provided name is removed. A Failure status is returned if the stack i ... *(Truncated)*
output	Success or Failure
input	[Function] CreateScope [Arguments] name(optional) [Description] Creates a new scope object and pushes it on top of the scope stack. If no name is provided the scope is anonymous and may be accessed only when it on the top of the scope stack. A Failur ... *(Truncated)*
output	Success or Failure
input	[Function] UpdateVariable [Arguments] variableName, newValue, scopeName(optional) [Description] Assigns a new value to the variable specified. If scopeName is not specified, the variable is accessed in the topmost scope on the stack. A Failure status ... *(Truncated)*
output	Success or Failure
Task 4 (9 examples)	
input	[Event Type] help [Action] reprompt [Audio Provided]
output	yes
input	[Event Type] noinput [Action] reprompt [Audio Provided]
output	no
input	[Event Type] exit [Action] exit interpreter [Audio Provided]
output	no

single_website_tables : mmo-champion.com	
Task 1 (15 examples)	
input	[Level] 384 [Type] Leather [Spec] Feral [Slot] Legs [Name]
output	Deep Earth Legguards
input	[Level] 384 [Type] Leather [Spec] Feral [Slot] Chest [Name]
output	Deep Earth Raiment
input	[Level] 384 [Type] Leather [Spec] Restoration [Slot] Shoulder [Name]
output	Deep Earth Mantle
Task 2 (23 examples)	
input	[Level] 384 [Type] Tier 13 [Slot] Token [Name] Crown of the Corrupted Protector [Instance] Dragon Soul [Boss] LFR Warmaster Blackhorn [Spec]
output	Armor
input	[Level] 384 [Type] Trinket [Slot] Trinket [Name] Bone-Link Fetish [Instance] Dragon Soul [Boss] LFR All Bosses Except Deathwing [Spec]
output	Melee
input	[Level] 384 [Type] Mace [Slot] Two-Hand [Name] Ataraxis, Cudgel of the Warmaster [Instance] Dragon Soul [Boss] LFR Warmaster Blackhorn [Spec]
output	Melee
Task 3 (12 examples)	
input	[ilvl] 85 [Type] Enchant [Item] Lesser Inscription of Charged Lodestone [Slot]
output	Shoulder
input	[ilvl] 346 [Type] Finger [Spec] Physical DPS [Item] Terrath's Signet of Balance [Slot]
output	Finger
input	[ilvl] 346 [Type] Finger [Spec] Melee [Item] Gorsik's Band of Shattering [Slot]
output	Finger
Task 4 (77 examples)	
input	[Level] 522 [Type] Mail [Spec] Physical DPS [Slot] Chest [Name] Carapace of Segmented Scale [Req. Standing]
output	Revered
input	[Level] 522 [Type] Leather [Spec] Physical DPS [Slot] Waist [Name] Darkfang Belt [Req. Standing]
output	Revered
input	[Level] 522 [Type] Trinket [Slot] Trinket [Name] Steadfast Talisman of the Shado-Pan Assault [Req. Standing]
output	Friendly

single_website_tables : studystack.com	
Task 1 (24 examples)	
input	[Answer] hard palte [Question]
output	The roof of the mouth is called the:
input	[Answer] middle ear [Question]
output	The malleus, incus, and stapes are located in the:
input	[Answer] Volar [Question]
output	The palm of the hand is called what?
Task 2 (15 examples)	
input	[Answer] Evert/eversion [Question]
output	Turning outward, typically used to describe ankle motion.
input	[Answer] Gliding motion [Question]
output	Occurs when one bone slides over another. EX. kneecap
input	[Answer] Invert/inversion [Question]
output	Turning inward, typically used to describe ankle motion,
Task 3 (13 examples)	
input	[Definition] freewriting, clustering, mapping, questioning, brainstorming [Term]
output	prewriting techniques.
input	[Definition] 5 senses, be specific, use comparisions, similes, metophores. Eliminate fluff words [Term]
output	good writing techniques
input	[Definition] (1) a topic and (2) a controlling idea [Term]
output	Two parts of a topic sentence
Task 4 (9 examples)	
input	[Definition] the amount of space something takes up [Term]
output	Mass
input	[Definition] a mixture made up of particles that are uniformly y distributed [Term]
output	homogeneous mixture
input	[Definition] the science of matter and how it changes [Term]
output	Chemistry

cluster_tables : 7	
Task 1 (7 examples)	
input	[Cookie Name] __utmb [Cookie Length] 30 minutes [Description]
output	Establish and continue a user session on the site
input	[Cookie Name] __utmz [Cookie Length] 6 months [Description]
output	Used to track traffic sources and page navigation
input	[Cookie Name] _UKWM [Cookie Length] 2 years [Description]
output	Used to identify traffic sources
Task 2 (8 examples)	
input	[Cookie Name or Service] MoodleSessionTest MoodleSession MoodleID_ [Purpose]
output	Our virtual learning environment, Moodle, uses cookies to record when visitors have successfully logged into the service.
input	[Cookie Name or Service] ASPSESSIONIDCQBSDQCQ [Purpose]
output	This is a functional cookie that does not contain any personal information and is automatically removed when the visitor closes their web browser.
input	[Cookie Name or Service] CAKEPHP [Purpose]
output	This is a functional cookie that does not contain any personal information and is automatically removed when the visitor closes their web browser.
Task 3 (9 examples)	
input	[Cookie] guest_id, ki [Information]
output	These cookies allow you to access the Twitter feed on the homepage.
input	[Cookie] use_hitbox [Information]
output	This is downloaded when you play an embedded YouTube video.
input	[Cookie] BX, localization [Information]
output	These cookies are downloaded by Flickr if you visit the page with the MEI Conference 2010 Photographs slideshow.
Task 4 (12 examples)	
input	[Cookie] pmx_cbtstat{ID} [Origin] www.whymsical.com [Persistency] Current session only [Information and Usage]
output	These cookies are set to records the expand/collapse state for a CBT Navigator block content.
input	[Cookie] pmx_YOfs [Origin] www.whymsical.com [Persistency] Page load time [Information and Usage]
output	This cookie will probably never see you. It is set on portal actions like click on a page number. The cookie is evaluated on load the desired page and then deleted. It is used to restore the vertical screen position as before the click.
input	[Cookie] AWNUTSWhymsicalcom [Origin] www.whymsical.com [Persistency] Expires according to user-chosen session duration [Information and Usage]
output	If you log-in as a member of this site, this cookie contains your user name, an encrypted hash of your password and the time you logged-in. It is used by the site software to ensure that features such as indicating new Forum and Private messages are ... *(Truncated)*

cluster_tables : 8	
Task 1 (7 examples)	
input	[0] Appearance [Scholarly Journals] Plain, “serious” cover Text with black & white graphs, charts, and photographs which ... *(Truncated)*
output	Generally glossy cover Color photographs and illustrations used to support the article as well as draw in readers
input	[0] Examples [Scholarly Journals] American Journal of Education Journal of the Evangelical Theological Society Modern Fiction Studies [Trade Journals]
output	Indiana Business Instrumentalist Preaching
input	[0] Validity [Scholarly Journals] Articles reviewed and evaluated by other experts in the field / discipline (peer reviewed / ... *(Truncated)*
output	Articles may be reviewed by one editor with knowledge related to the topic
Task 2 (15 examples)	
input	[DATABASE TITLE] Engineered Materials Abstracts [FULL DESCRIPTION] Comprehensive index to world literature on engineered ... *(Truncated)*
output	no
input	[DATABASE TITLE] Engineering Research Database [FULL DESCRIPTION] The ProQuest Engineering Research Database covers the ... *(Truncated)*
output	no
input	[DATABASE TITLE] ENGnetBASE [FULL DESCRIPTION] The ENGnetBase eBook collection includes over 2300 cutting-edge and bestselling ... *(Truncated)*
output	yes
Task 3 (20 examples)	
input	[Access] Website [2] Choose My Plate The new food and dietary guidelines! Also included are related links such as: farmer’s markets, nutrition labels and food safety. Created by the USDA. [Subject]
output	Health & Nutrition
input	[Access] Website [2] Library of Congress; Performing Arts Encyclopedia This is an amzing guide to the performing arts. You can ... *(Truncated)*
output	Art
input	[Access] Library Card Required [2] Encyclopedia Britannica This encyclopedia has A LOT of information, which is great, but ... *(Truncated)*
output	Cultures
Task 4 (6 examples)	
input	[Time Frame of Event] Seconds/minutes/hours Provides sketchy details, may be inaccurate but good for firsthand accounts [Information Resource]
output	Television/radio/internet
input	[Time Frame of Event] Six months or more In depth analysis of event written by experts in their field. In most cases, ... *(Truncated)*
output	Scholarly Journals
input	[Time Frame of Event] Next day or two More details and greater accuracy, the first rough draft of history [Information Resource]
output	Newspapers

cluster_tables : -1	
Task 1 (7 examples)	
input	[Domain Name] TinyHomeForSale.com [Price] $1,999 [Buy] Buy it Now [Keyword]
output	Tiny Home For Sale
input	[Domain Name] DomainSalesHistory.com [Price] Offer [Buy] Buy it Now [Keyword]
output	Domain Sales History
input	[Domain Name] NearbyForSale.com [Price] $999 [Buy] Buy it Now [Keyword]
output	Nearby For Sale
Task 2 (8 examples)	
input	[You are...] Supportive [You should have...]
output	A strong stomach
input	[You are...] Dependable [You should have...]
output	Good ethical standards
input	[You are...] Organized [You should have...]
output	Excellent attention to detail
Task 3 (10 examples)	
input	[Indonesian] perangko [English]
output	stamp
input	[Indonesian] surat [English]
output	letter
input	[Indonesian] terdaftar [English]
output	registered mail
Task 4 (9 examples)	
input	[Endpoint/Outcome Measure] Vertebral Morphometry (6-point, 95-point) [Modality] X-Ray, DXA, CT [Description]
output	Automatic identification of vertebral body margins
input	[Endpoint/Outcome Measure] Microarchitecture [Modality] MRI, High resolution QCT (HR-pQCT) [Description]
output	Measurement of trabecular and cortical bone microarchitecture
input	[Endpoint/Outcome Measure] Bone Marrow Edema (BME) [Modality] X-Ray, MRI [Description]
output	Detection of pathogenic changes in the bone marrow of the femoral head

cluster_tables : 3	
Task 1 (25 examples)	
input	[COOKIE name] CATEGORY_INFO [COOKIE Description]
output	Stores the category info on the page, that allows to display pages more quickly.
input	[COOKIE name] FRONTEND [COOKIE Description]
output	You sesssion ID on the server.
input	[COOKIE name] CART [COOKIE Description]
output	The association with your shopping cart.
Task 2 (25 examples)	
input	[COOKIE name] WISHLIST_CNT [COOKIE Description]
output	The number of items in your Wishlist.
input	[COOKIE name] NO_CACHE [COOKIE Description]
output	Indicates whether it is allowed to use cache.
input	[COOKIE name] GUEST-VIEW [COOKIE Description]
output	Allows guests to edit their orders.
Task 3 (25 examples)	
input	[COOKIE name] CUSTOMER_AUTH [COOKIE Description]
output	An indicator if you are currently logged into the store.
input	[COOKIE name] CUSTOMER [COOKIE Description]
output	An encrypted version of your customer id with the store.
input	[COOKIE name] STORE [COOKIE Description]
output	The store view or language you have selected.
Task 4 (25 examples)	
input	[COOKIE name] NO_CACHE [COOKIE Description]
output	Indicates whether it is allowed to use cache.
input	[COOKIE name] LAST_CATEGORY [COOKIE Description]
output	The last category you visited.
input	[COOKIE name] POLL [COOKIE Description]
output	The ID of any polls you have recently voted in.

nlp_train	
numer_sense (100 examples)	
input	All scorpions have an additional [MASK] segments after the initial seven, ending in a sharp sting.
output	five
input	Heart failure affects about [MASK] million people in the United States.
output	five
input	Ribosomes have [MASK] subunits - small and large.
output	two
spider (100 examples)	
input	What are the names of the climbers, ordered by points descending?
output	SELECT Name FROM climber ORDER BY Points DESC
input	Find the first names and offices of all instructors who have taught some course and also find the course description.
output	SELECT T2.emp_fname , T4.prof_office , T3.crs_description FROM CLASS AS T1 JOIN employee AS T2 ON T1.prof_num = T2.emp_num JOIN course AS T3 ON T1.crs_code = T3.crs_code JOIN professor AS T4 ON T2.emp_num = T4.emp_num
input	What is the county that produces the most wines scoring higher than 90?
output	SELECT T1.County FROM APPELLATIONS AS T1 JOIN WINE AS T2 ON T1.Appelation = T2.Appelation WHERE T2.Score > 90 GROUP BY T1.County ORDER BY count(*) DESC LIMIT 1
yahoo_answers_topics (100 examples)	
input	question_title: man date women but has serious secret interest exclusively in men who are women from waist up? [SEP] question_content: and who wear make-up etc - is he really interested in men, and too afraid to come out of the closet or what? [SEP ... *(Truncated)*
output	Society & Culture
input	question_title: bungee jumping site in victoria??? [SEP] question_content: i am trying to find a site for bungee jumping around melbourne. i went thru the internet but couldnt find much. can anyone give me some info pls coz i ve been dreaming for t ... *(Truncated)*
output	Sports
input	question_title: celebs criminal conviction? [SEP] question_content: can anybody suggesting some famous celebs or successful persons who's got criminal conviction? [SEP] best_answer: Lots of celebrity activists have had criminal convictions, usuall ... *(Truncated)*
output	Politics & Government
piqa (100 examples)	
input	goal: Preserve expensive lipstick. [SEP] solution 1Keep in clothes drawer. [SEP] solution 2Keep in fridge.
output	1
input	goal: How to wash a dog. [SEP] solution 1Wet the dog with warm water, apply shampoo, lather and massage into fur, no need to rinse out all shampoo. Repeat process with conditioner if desired. [SEP] solution 2Wet the dog with warm water, apply shampoo ... *(Truncated)*
output	1
input	goal: To add a light inside a lamp. [SEP] solution 1Get wire with a plug, and chain, and feed the chain on. Then put on a washer -this should be decently big, and this is how the shade part will be attached. Then tape the wire to the socket, and scre ... *(Truncated)*
output	1

nlp_test	
ag_news (100 examples)	
input	Delegation Is Delayed Before Reaching Najaf AGHDAD, Iraq, Aug. 17 A delegation of Iraqis was delayed for security reasons today but still intended to visit Najaf to try to convince a rebellious Shiite cleric and his militia to evacuate a shrine in t ... *(Truncated)*
output	World
input	Restive Maldives eases curfew after rounding up dissidents (AFP) AFP - A curfew in the capital of the Maldives was eased but parliament sessions were put off indefinitely and emergency rule continued following last week's riots, officials and residen ... *(Truncated)*
output	World
input	Another Major Non-Factor Another major, another disappointment for Tiger Woods, the No. 1 ranked player in the world who has not won a major championship since his triumph at the 2002 U.S. Open.
output	Sports
amazon_polarity (100 examples)	
input	title: Prompt shipment [SEP] content: I still haven't had time to watch the video to comment about the quality, but it was shipped promptly and seems to be in good order.
output	positive
input	title: Hey, we gotta talk [SEP] content: well, i gotta say this is one of her better albums. I'm real is da bomb and so is the I'm real (murder remix) she and ja rule sound SOOOOOO fine together. Love dont' cost a thing is hott too but Play is almost ... *(Truncated)*
output	positive
input	title: absolute lemon [SEP] content: I probably have as much experience with 11x17 capable color printers as anyone in the world and I've got to say this is easily the most difficult and unsatisfactory printer I have ever dealt with. HP's last 11x17 ... *(Truncated)*
output	negative
commonsense_qa (100 examples)	
input	What is the main purpose of farmers?
output	supply food
input	When drinking booze what can you do to stay busy?
output	examine thing
input	If you are prone to postpone work what will you have to do in order to finish on time?
output	hasten
dbpedia_14 (100 examples)	
input	Edward Albert Eddie Nicholls (born 10 December 1947 in British Guiana) is an international Test and One Day International cricket umpire. Nicholls officiated in 17 Tests from 1997 to 2001 and 46 ODIs from 1995 to 2005. Nicholls is one of four West In ... *(Truncated)*
output	Athlete
input	Cryptolepis is a plant genus in the family Apocynaceae. It includes some 42 species.
output	Plant
input	Mike Enneking is a retired American soccer player who played professionally in the Western Soccer Alliance and Major Indoor Soccer League. He is the Vice President of Crossfire Premier Soccer Club.Enneking graduated from Bellarmine Preparatory School ... *(Truncated)*
output	Athlete

nlp_test	
ethos-race (87 examples)	
input	In my dream I wished you have died! Don't call yourself man, while talking like queen.
output	false
input	Is that your nose or is that Thor's Hammer?
output	false
input	All girls like you should have a pimp to guide them to successful prostitution
output	false
financial_phrasebank (100 examples)	
input	Profit before taxes was EUR 5.4 mn , up from EUR 3.6 mn a year earlier .
output	positive
input	The order was valued at USD12 .2 m.
output	neutral
input	The company expects net sales to significantly increase from 2009 .
output	positive
qasc (100 examples)	
input	what is tourette syndrome?
output	trait
input	Animals that are _ provide little if any care to their young.
output	cold blooded
input	What can be used for transportation?
output	trailers and boats
sciq (100 examples)	
input	All alkaline Earth metals have similar properties because they all have two valence electrons. They readily give up their two valence electrons to achieve a full outer energy level, which is the most stable arrangement of electrons. As a result, the ... *(Truncated)*
output	valence electrons
input	Exposure gives an indication of the amount of radiation that travels through the air. Two factors influence the amount of exposure a person may receive – time and intensity. Acute exposure indicates a large amount of radiation received over a short ... *(Truncated)*
output	chronic exposure
input	Ventricular Systole Ventricular systole (see Figure 19.27) follows the depolarization of the ventricles and is represented by the QRS complex in the ECG. It may be conveniently divided into two phases, lasting a total of 270 ms. At the end of atrial ... *(Truncated)*
output	pulmonary and aortic semilunar

nlp_test	
tweet_eval-stance_atheism (52 examples)	
input	The worst day of my life so far is here, setting my Nan to rest. Even as a physicist, times like these make you wonder. #SemST
output	none
input	I will dwell in a peaceful habitation, in secure dwellings, and in quiet resting places -Isa. 32:18 #SemST
output	against
input	@user sweet! Congratulations to a rational decision. #SemST
output	none
yelp_polarity (100 examples)	
input	Very disappointed in this salon. Set an appt 4 days ahead of time. Area were I for my set put on was dirty from a past client. The mail tech did not talk, I felt rushed through my appt which resulted in me leaving unhappy. I won't be returning.
output	negative
input	Our flight arrived to Vegas earlier than excepted, so we expected our room not to be ready. When we arrived at the hotel on May 19th, the front desk girl offered us a room that was ready on the 28th floor that wasn't facing the Bellagio fountain. I b ... *(Truncated)*
output	positive
input	My poor children who live out of state, have no idea how cheap and ugly the flowers I just received from Carmel Florist are. They do not resemble the online photo at all. I actually laughed at the gentleman who delivered them to my door. They spent ... *(Truncated)*
output	negative

ACL 2023 Responsible NLP Checklist

A For every submission:

☐ A1. Did you describe the limitations of your work?
Left blank.

☐ A2. Did you discuss any potential risks of your work?
Left blank.

☐ A3. Do the abstract and introduction summarize the paper's main claims?
Left blank.

☐ A4. Have you used AI writing assistants when working on this paper?
Left blank.

B ☐ Did you use or create scientific artifacts?

Left blank.

☐ B1. Did you cite the creators of artifacts you used?
Left blank.

☐ B2. Did you discuss the license or terms for use and / or distribution of any artifacts?
Left blank.

☐ B3. Did you discuss if your use of existing artifact(s) was consistent with their intended use, provided that it was specified? For the artifacts you create, do you specify intended use and whether that is compatible with the original access conditions (in particular, derivatives of data accessed for research purposes should not be used outside of research contexts)?
Left blank.

☐ B4. Did you discuss the steps taken to check whether the data that was collected / used contains any information that names or uniquely identifies individual people or offensive content, and the steps taken to protect / anonymize it?
Left blank.

☐ B5. Did you provide documentation of the artifacts, e.g., coverage of domains, languages, and linguistic phenomena, demographic groups represented, etc.?
Left blank.

☐ B6. Did you report relevant statistics like the number of examples, details of train / test / dev splits, etc. for the data that you used / created? Even for commonly-used benchmark datasets, include the number of examples in train / validation / test splits, as these provide necessary context for a reader to understand experimental results. For example, small differences in accuracy on large test sets may be significant, while on small test sets they may not be.
Left blank.

C ☐ Did you run computational experiments?

Left blank.

☐ C1. Did you report the number of parameters in the models used, the total computational budget (e.g., GPU hours), and computing infrastructure used?
Left blank.

The Responsible NLP Checklist used at ACL 2023 is adopted from NAACL 2022, with the addition of a question on AI writing assistance.

☐ C2. Did you discuss the experimental setup, including hyperparameter search and best-found hyperparameter values?
Left blank.

☐ C3. Did you report descriptive statistics about your results (e.g., error bars around results, summary statistics from sets of experiments), and is it transparent whether you are reporting the max, mean, etc. or just a single run?
Left blank.

☐ C4. If you used existing packages (e.g., for preprocessing, for normalization, or for evaluation), did you report the implementation, model, and parameter settings used (e.g., NLTK, Spacy, ROUGE, etc.)?
Left blank.

D ☐ Did you use human annotators (e.g., crowdworkers) or research with human participants?

Left blank.

☐ D1. Did you report the full text of instructions given to participants, including e.g., screenshots, disclaimers of any risks to participants or annotators, etc.?
Left blank.

☐ D2. Did you report information about how you recruited (e.g., crowdsourcing platform, students) and paid participants, and discuss if such payment is adequate given the participants' demographic (e.g., country of residence)?
Left blank.

☐ D3. Did you discuss whether and how consent was obtained from people whose data you're using/curating? For example, if you collected data via crowdsourcing, did your instructions to crowdworkers explain how the data would be used?
Left blank.

☐ D4. Was the data collection protocol approved (or determined exempt) by an ethics review board?
Left blank.

☐ D5. Did you report the basic demographic and geographic characteristics of the annotator population that is the source of the data?
Left blank.